

AGK 7

7.1.2

→ The program produces the same output as in class (1 error)

IRIS in 1-NN

Manhattan: 143 / 150 correct (95.3%)

Euclidean: 144 / 150 correct (96%)

If random

↓

33% (numpy)

Manhattan

No. of tests	10	20	30	40	80	160	320
Accuracy	40%	40%	50%	62,5%	73,8%	80%	85%

Euclidean

No. of tests	10	20	30	40	80	160	320
Accuracy	50%	40%	50%	62,5%	75%	81,3%	85,6%

With more data the algorithm performs better
And euclidean distance performs slightly better

7.2

IRIS, for $k=3$

For both distances = 144 / 150 (96%) → improved Manhattan

Manhattan (MNIST)

No of Tests	20	20	20	80	80	80	160	160	160	320	320	320	320
k	3	5	10	3	5	10	3	5	10	3	5	10	20
Accuracy	20%	20%	20%	66,25%	62,5%	53,75%	77,5%	75,6%	70%	81,25%	80,3%	78,1%	70%

Without weighting the distance, doesn't appear improvement on the algorithm, specially for a bigger k .